

PILATUS, Switzerland

WMO: 066590

Lat: 46.9789N

Lon: 8.2522E

Elev: 6923

StdP: 11.37

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.7	6.3	-23.6	1.9	28.3	-16.7	2.9	25.5	45.1	28.8	39.5	25.0	14.1	40	0.853

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	65.6	54.3	63.0	53.2	60.6	52.0	56.9	62.5	55.3	60.4	53.7	58.6	8.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.7	82.7	59.5	53.0	77.6	57.9	51.5	73.2	56.5	27.6	62.8	26.4	60.6	25.2	58.8	63.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.4	31.1	26.6	DB	-0.5	69.4	5.2	3.1	-4.3	71.6	-7.3	73.4	-10.2	75.1	-14.0	77.4	(4)
(5)			WB	-1.6	60.4	4.7	1.7	-5.0	61.7	-7.8	62.7	-10.4	63.6	-13.9	64.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	24.4	23.6	27.9	33.7	39.0	46.9	49.7	49.8	44.0	40.4	32.1	27.5	(6)
(7)		DBStd	12.40	7.99	9.67	8.77	8.01	7.96	8.09	7.16	7.56	7.51	8.56	9.00	8.48	(7)
(8)		HDD50	5155	795	739	684	489	352	151	95	101	206	308	537	698	(8)
(9)		HDD65	10346	1260	1159	1149	939	807	542	475	472	631	762	987	1163	(9)
(10)		CDD50	284	0	0	0	0	10	59	86	94	25	10	0	0	(10)
(11)		CDD65	2	0	0	0	0	0	1	1	0	0	0	0	0	(11)
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	12.1	13.6	13.6	12.7	12.1	11.6	10.4	9.9	9.2	11.0	12.7	14.0	14.4	(14)
(15)	Precipitation	PrecAvg	64.8	3.7	4.1	4.3	5.0	6.9	7.2	7.2	7.4	5.4	4.2	4.9	4.6	(15)
(16)		PrecMax	81.5	8.8	12.1	8.5	10.0	10.9	11.1	13.7	16.0	9.2	7.5	13.6	11.4	(16)
(17)		PrecMin	49.3	0.2	1.3	1.7	1.1	1.7	4.3	3.1	2.2	2.1	0.2	0.2	0.2	(17)
(18)		PrecStd	7.3	2.2	2.4	1.9	2.0	2.8	1.7	2.6	2.9	1.8	1.8	2.7	2.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.3	46.7	48.4	52.7	60.5	69.4	68.7	68.5	61.4	59.5	52.1	46.8	(19)
(20)			MCWB	32.2	32.5	35.2	41.7	49.6	55.6	56.7	54.7	49.5	45.3	37.7	32.4	(20)
(21)		2%	DB	41.0	43.5	44.5	49.6	56.0	64.5	64.8	65.4	58.3	55.8	48.9	43.7	(21)
(22)			MCWB	30.1	30.5	33.6	40.6	47.2	53.4	54.5	54.0	48.6	43.2	37.0	30.5	(22)
(23)		5%	DB	37.8	39.1	41.8	47.0	53.3	61.1	62.1	62.8	56.0	53.3	46.2	40.9	(23)
(24)			MCWB	29.0	29.5	31.8	38.9	45.5	52.7	53.4	53.1	47.8	42.2	35.7	30.1	(24)
(25)		10%	DB	34.9	35.5	39.0	44.4	50.4	58.1	59.6	60.3	53.9	50.9	43.9	38.6	(25)
(26)			MCWB	28.2	27.5	31.0	37.1	43.9	51.0	52.2	52.0	46.7	40.8	34.4	29.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.2	35.5	38.3	44.8	52.3	59.1	59.6	59.0	53.9	48.4	41.9	36.6	(27)
(28)			MCD B	39.5	42.1	43.5	49.9	58.1	65.6	66.2	63.9	58.0	53.8	47.9	40.0	(28)
(29)		2%	WB	33.6	32.7	36.0	42.2	49.2	56.2	56.6	56.7	51.7	46.1	39.4	34.2	(29)
(30)			MCD B	36.3	37.6	40.6	47.6	54.0	61.6	62.1	62.0	55.6	51.5	44.6	37.9	(30)
(31)		5%	WB	31.7	30.9	34.2	40.0	46.5	54.2	54.9	55.1	49.8	44.5	37.7	32.4	(31)
(32)			MCD B	34.7	36.1	38.9	45.3	51.8	59.1	59.9	60.3	53.8	50.4	43.0	37.2	(32)
(33)		10%	WB	29.5	29.1	32.4	38.1	44.4	52.0	53.3	53.2	48.0	43.0	35.9	31.0	(33)
(34)			MCD B	33.5	34.8	37.0	43.3	49.0	56.6	57.9	58.3	52.3	48.5	42.0	36.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	9.2	9.3	9.1	9.0	9.0	9.2	8.8	8.6	9.3	9.1	9.2	(35)
(36)			MCDBR	10.6	11.2	11.1	10.6	11.3	10.8	11.2	10.8	10.0	10.8	10.5	10.6	(36)
(37)			MCWBR	7.0	7.2	7.8	8.8	10.1	10.5	9.9	10.4	9.9	8.0	6.6	6.6	(37)
(38)		5% WB	MCD BR	10.1	10.4	9.7	10.4	10.8	10.1	10.4	9.8	9.1	9.7	9.7	10.3	(38)
(39)			MCWBR	9.2	8.2	7.6	8.9	9.8	10.5	10.0	10.3	9.5	7.9	7.5	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.227	0.247	0.278	0.307	0.320	0.324	0.325	0.318	0.298	0.279	0.250	0.225	(40)	
(41)		taud	2.452	2.420	2.423	2.415	2.424	2.442	2.423	2.460	2.512	2.555	2.550	2.469	(41)	
(42)		Ebn at Noon	287	300	303	302	299	298	296	294	292	284	275	275	(42)	
(43)		Edh at Noon	20	26	30	34	35	35	35	32	28	23	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	427	711	1092	1392	1558	1820	1783	1551	1189	767	444	352	(44)	
(45)		RadStd	51	93	155	211	153	167	158	148	116	84	60	52	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (119 neighbors)	N/A	N/A	N/A	+1.30	+0.72	+0.56	N/A	N/A	+103	+27	

Nomenclature: See separate page